

Health Analytics for a Home Health Startup

INFO Graduate Data Analytics Final Project

Individual or Teams of Two

Project Overview

A home health care startup is evaluating where to expand operations in the United States. The company wants to identify which states have populations with the highest potential demand for home health services and understand which types of individuals are most likely to require elevated health support.

Your team will act as data analysts hired to explore this question using real public health data from the **Behavioral Risk Factor Surveillance System (BRFSS)**, a large national health survey conducted by the Centers for Disease Control and Prevention (CDC). The dataset contains responses from hundreds of thousands of individuals and includes information about demographics, health status, chronic conditions, and health behaviors.

This project is organized into four sequential phases. Two phases include instructor checkpoints that must be completed before proceeding.

Learning Objectives

By completing this project, you will:

- Work with a large real-world public health dataset
- Perform exploratory data analysis and communicate findings clearly
- Design and justify a proxy target variable
- Apply clustering to identify population segments
- Build and evaluate a classification model
- Translate analytical findings into a business recommendation

Dataset

You will work with a curated subset of the CDC Behavioral Risk Factor Surveillance System (BRFSS). BRFSS is an annual public health survey that collects information on health conditions, health behaviors, and demographic characteristics of U.S. adults.

Each row in the dataset represents one survey respondent. The dataset includes hundreds of thousands of observations collected from all U.S. states.

The variables provided fall into four general categories:

- **Demographics:** age group, sex, education level, income group, race category, employment status, and state of residence.

- **Health Status Indicators:** self-reported general health, number of days of poor physical health, number of days of poor mental health, and number of days when poor health limited normal activities.
- **Health Behaviors and Risk Factors:** smoking status, smoking frequency, body mass index category, and exercise participation.
- **Chronic Health Conditions:** indicators for conditions such as diabetes, heart disease, stroke, and chronic obstructive pulmonary disease (COPD).

A data dictionary describing all variables and survey codes will be provided.

Important: BRFSS is a survey dataset, and many variables use numeric codes for responses such as “Don’t know” or “Refused.” For example, values such as 7, 9, 77, and 99 frequently represent non-substantive responses rather than meaningful measurements. During data cleaning you should review the data dictionary and determine how these values should be handled.

Project Structure

The project consists of four phases that mirror a typical analytics workflow.

Phase 1: Data Cleaning and Exploratory Analysis

Task 1: Data Cleaning

Prepare the dataset for analysis. Your cleaning process should include:

- Handling missing or invalid survey responses
- Converting categorical survey codes into usable formats
- Selecting and justifying the variables you will carry forward
- Documenting each major data preparation decision

Your notebook should clearly explain the reasoning behind your cleaning choices.

Task 2: Exploratory Data Analysis

Conduct exploratory analysis to understand the structure and distribution of the data. Your analysis must include at least four visualizations.

Possible examples include:

- Distribution of key health indicators
- Health outcomes by age group or income level
- Comparison of chronic condition prevalence across states
- Relationships between lifestyle behaviors and health outcomes

Each visualization should include a short interpretation explaining what the result suggests.

Checkpoint 1: Target Variable Proposal Due 1 APR

After completing your exploratory analysis, submit a short written brief (approximately one half page) proposing your target variable, `HighSupportNeed`.

Your proposal should:

- Identify the BRFSS variables you plan to combine
- Explain why those variables represent indicators of elevated home health support need
- Describe the logic used to construct the binary variable

You may not proceed to Phase 2 until the instructor has approved this proposal.

Phase 2: Target Variable Construction and Clustering

Task 3: Construct the Target Variable

Using the approved variables from your proposal, construct the binary variable `HighSupportNeed`.

Your notebook must:

- Implement the logic described in your proposal
- Report the percentage of respondents classified as high support need
- Discuss whether the resulting classes are balanced or imbalanced

Task 4: Clustering Analysis

Use a clustering method (such as K-Prototype) to identify population segments within the dataset.

Your clustering analysis should include:

- Justification for the variables used in clustering
- A method for selecting the number of clusters (for example the elbow method or silhouette scores)
- Summary statistics describing each cluster
- Descriptive names for the clusters
- A plain-language interpretation of what each cluster represents

Add cluster membership as a new column in your dataset. This feature will be used during classification modeling.

Checkpoint 2: Cluster Review - Due 15 APR

Submit a short summary describing the clusters discovered in your analysis. For each cluster, provide:

- A descriptive name
- Key characteristics of the cluster
- Your interpretation of the type of individual represented

You should also discuss whether cluster membership appears likely to be predictive of `HighSupportNeed`.

Phase 3: Classification Modeling

Task 5: Predict High Support Need

Build a classification model that predicts `HighSupportNeed`. Acceptable models include Logistic Regression, Decision Tree, Random Forest, Boosted Trees, or ANN.

Your model must include cluster membership as one of the input features.

When selecting features for your classification model, consider whether any variables used to construct the target variable may create unintended leakage or trivial prediction. Explain how you addressed this issue in your modeling design.

Your evaluation must include:

- A train-test split
- Accuracy and a confusion matrix
- Precision/Recall
- A discussion of class imbalance and any strategies used to address it. Possible approaches include adjusting the classification threshold, using class weights, or applying a resampling technique such as oversampling or SMOTE. If you choose to apply one of these methods, explain why it was appropriate for your model.
- Feature importance or model coefficients

You should also compare your model against a simple baseline prediction.

Because cluster membership is derived from the data, you should consider how it fits within your modeling workflow. In particular, you should avoid allowing information from the test set to influence the clustering process. Briefly explain how your workflow addresses this issue and how cluster membership is incorporated into the modeling pipeline.

Phase 4: State Analysis and Business Recommendation

Task 6: State-Level Analysis

Examine differences across states. Your analysis should consider more than raw counts.

Examples include:

- Prevalence of `HighSupportNeed` by state
- Which clusters are most common in high-need states
- Demographic or behavioral differences between states

Identify the state that appears to represent the best expansion opportunity for the startup.

Task 7: Business Recommendation

Conclude the project with a 10-15 minute presentation.

Where should the home health startup consider expanding first, and which population segments appear most likely to require elevated health support?

Your recommendation should reference evidence from your exploratory analysis, clustering results, model findings, and state-level analysis.

You should also acknowledge any limitations of your approach.

Final Deliverables

- A Python notebook (.ipynb) containing all code and explanations
- Visualizations embedded within the notebook
- Target variable proposal (Checkpoint 1)
- Cluster summary (Checkpoint 2)
- Final Presentation and business recommendation

Your notebook should read as a coherent analytical report rather than a sequence of code cells.

Grading Expectations

This is a graduate-level project. Evaluation will focus not only on whether your code executes, but also on the quality of your reasoning.

Strong projects will:

- Justify analytical decisions clearly
- Recognize limitations of proxy variables

- Interpret results in plain language
- Connect analytical findings to business implications

Project Rubric

Component	Points	Description
Phase 1: Data Preparation and Exploration		
Data Cleaning and Preparation	10	Dataset is prepared appropriately. Missing values, special survey codes, and categorical encodings handled correctly. Data preparation decisions are clearly documented and justified.
Exploratory Data Analysis	10	Visualizations are clear and relevant. Analysis explores relationships between health indicators, demographics, and behaviors. Interpretations explain what patterns suggest about health support needs.
Checkpoint 1: Target Variable Proposal	10	Target variable proposal clearly identifies variables used to construct HighSupportNeed . Justification demonstrates logical reasoning and understanding of the dataset.
Phase 2: Target Construction and Clustering		
Target Variable Construction	10	Binary target variable implemented correctly. Class balance is reported and discussed. Logic follows the approved proposal.
Clustering Analysis	10	Appropriate clustering method selected. Number of clusters justified. Clusters are interpreted clearly with meaningful descriptions and supporting statistics.
Checkpoint 2: Cluster Review	10	Cluster summaries clearly describe population segments. Interpretations demonstrate thoughtful reasoning about the characteristics of each cluster.
Phase 3: Predictive Modeling		
Classification Model Development	10	Model built correctly with appropriate feature selection and train-test split. Includes cluster membership as a feature. Modeling workflow is logically constructed.
Model Evaluation and Interpretation	10	Evaluation includes accuracy, confusion matrix, and precision/recall. Discussion addresses class imbalance and model limitations. Feature importance or coefficients are interpreted meaningfully.
Phase 4: State Analysis and Business Recommendation		
State-Level Analysis	10	State comparisons move beyond simple counts and provide meaningful insights about geographic variation in potential demand for home health services.
Final Presentation	20	Presentation is professional, organized, and well prepared. Students clearly explain the analytical process, findings, and business recommendation. Visuals are clear and presenters demonstrate strong understanding of their work.
Total	100	